



First Step on the Moon

Find out why Neil Armstrong was the one who set his foot first on the moon
<http://dgit.in/MoonFirstStep>



DARPA builds own SoC

DARPA's ELASTx Microchip is a complete, all-on-one chip system that can boost power of transmitted signals <http://dgit.in/DARPASoC>

beyond the extent of our knowledge. As a highly seemingly *advanced* species we have imagined numerous practical means to redesign planets within a few lifetimes. Already, scientists have begun experimenting with the various approaches via which this goal can be accomplished. By embracing terraforming, especially within the local neighbourhood, we can apply our ever increasing database of knowledge about planets and moons to determine where we will thrive.

Colonisation is an essential part of the process, as it would require vast resources both technological and labour-oriented to make terraforming feasible. Projects such as the Biosphere Dome experiment have revealed invaluable lessons over the last 30 years. The ability to build, develop and sustain organic life in a myriad of environmental conditions is a basic necessity of colonisation. In addition to which, the technology necessary to build durable and secure life support systems has long been established and is on a constant upward state of development as we innovate new construction materials. But colonisation is just the interim step which humanity needs to prepare for as it transitions to rebuilding a whole planet suited to its needs.

Terra Incognita Exotica

Long before humanity had ever stepped into space scientists were intrigued by the idea of creating another Earth. In 1961, famed astronomer Carl Sagan published his idea of Venus being a candidate in *Science* magazine. The means of doing so were elegant and naturalistic. Sagan opined that by seeding the atmosphere with algae we could convert the nitrogen and carbon dioxide into life-sustaining compounds. The resulting transformation would reduce the naturally high greenhouse effects on Venus and lower temperatures within human-tolerance levels. Of course, further research into Venus's composition established that Sagan's methodology would prove useless since the planet is enveloped in thick clouds of concentrated sulphuric acid under high atmospheric pressure.

Carl Sagan even pioneered the idea of Mars as a potential candidate in 1973 in an



As our closest neighbour, Mars is the easiest planet to reach and re-imagine as a second Earth.

article published in the journal *Icarus*. But it wasn't until 1982 that the term "terraforming" joined the popular lexicon as a replacement for "planetary engineering" and "planetary ecosynthesis". Terraforming as it exists today requires the presence of a planet, moon or other heavenly body which at the very minimum is capable of hosting liquid water, conditions necessary to assemble complex organic life and a sustained source of energy for metabolic activities. Beyond these broad parameters the real challenge is creating technology that can foster these essential traits.

Each candidate planet comes with unique challenges to terraforming depending on a multitude of factors. Venus is relatively a simpler problem since its three key transformation criterias are inter-related. The composition of Venus makes it an extremely hot, atmospherically dense and toxic environment. The reigning temperatures can reach over 450 degrees Celsius with an unbreathable atmosphere that is over 90 times as dense as Earth, consisting of carbon dioxide and sulphur dioxide. In addition to these challenges, the planet also receives twice the sunlight and reducing the insolation would greatly ease surface temperatures. Other options such as Jupiter's moon Europa, our own Moon and Saturn's moon Titan, are all possible candidates for terraforming but nothing compares to Mars as the most viable option.

Making Mars Home?

Mars on the other hand is a completely different problem. The nearly barren planet is farther away from Earth with significantly lesser gravity. The lack of a dense atmosphere and a magnetosphere also renders it prone to radiation threats from the Sun. With only a thin carbon dioxide atmosphere sustaining organic life is impossible. The most recent exploratory results from the Mars Curiosity rover tell us that the only remaining source of water on the planet is in the form of permafrost under its polar surfaces and within its soil which could prove to be a silver lining.

On the whole the challenge of terraforming Mars would still be more preferable than Venus as it is only on the fringes of the habitable zone orbital range of the